SOFTWARE RECRUITMENT TASKS

Prerequisite : Dual boot your system with **Ubuntu 22.04 LTS**, Install **ROS2 Humble**

- **Task 1: Learn basic ROS2 (ROS Humble), Linux (Ubuntu) and some basic Git commands**

  For ROS2 Humble, refer to the documentation https://docs.ros.org/en/humble/index.html For Git, refer to the documentation at https://www.w3schools.com/git/default.asp

- **Task 2: ROS2 Sensor Data Communication and Visualization**

  The rover is equipped with three onboard sensors. Implement a Publisher-Subscriber pair using ROS2 Humble to transmit and receive sensor data in an array format. Assume sensor data to be a float array[3]. Plot the received data in real time using Matplotlib or any other graphing library.

- **Task 3: Object Detection with Aruco markers**

  Develop an object detection software to identify Aruco markers. You may use any suitable library for detection. Provide a video recording or image of the functioning software. Sample Aruco markers

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- **Task 4: Pseudocode for autonomous navigation**
  You are provided with an autonomous rover operating on Martian terrain. The rover is equipped with a robotic arm, GPS for navigation, and a depth camera capable of detecting and scanning objects and cones within a range of up to 15 meters. The objective of the rover is to autonomously locate, pick up, and transport objects to their designated destinations. Key conditions:

  - The operational area is limited such that no object is located more than 500 meters from the rover's starting position. The terrain contains three pairs of colored cones: red, blue, and green. Each pair consists of:
    - One cone marking the object pickup location .
    - The corresponding cone marking the delivery location
    - Objects placed at a cone of a specific color must be delivered to the matching cone of the same color.

  Design a clear and structured pseudo code that enables the rover to pickup objects using the robotic arm , and deliver each object autonomously to its destination cone .

Bonus Tasks

Soil Classification Model:

Train a **classification model** to detect soil types using the provided dataset. Plot the results of the model (accuracy, loss, etc.).

Dataset Link